Scientific Computing Exercise Set 3

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***Repository**: (Scientific Computing) https://github.com/JIA0501/Scientific-Computing*

I. Introduction

In this report, numerical methods are applied to study two distinct physical problems: fluid flow past a cylinder and electromagnetic wave propagation in an indoor environment. Although the underlying physics differ, both problems can be formulated in terms of partial differential equations and solved using computational techniques.

The objective of the first part is to understand how *numerical modelling choices influence the behaviour of a simulation*. We consider incompressible flow past a cylinder, a classical test case exhibiting a Kármán vortex street, because it provides a sensitive probe of the balance between convection, diffusion, and numerical stability.

Three fundamentally different discretization philosophies were investigated:

- Finite Difference (FD): explicit control over every discrete operator,
- Finite Element Method (FEM): variational formulation using NGSolve,
- Lattice Boltzmann Method (LBM): mesoscopic transport model.

The central question guiding this first part was:

> *How does the way we discretize the equations change what solution we obtain, how expensive it is to compute, and how reliable it remains as the Reynolds number increases?*

In contrast, Challenge 3.2 considers the propagation of electromagnetic waves inside a building. The goal is to model the spatial distribution of signal strength generated by a wireless router and to determine an optimal placement that maximizes coverage. This is achieved by solving the Helmholtz equation and evaluating the resulting field at selected measurement locations.

Despite the differences in physical context, both challenges follow a similar computational approach. A governing equation is discretized and solved numerically, and the resulting solution is post-processed to extract meaningful physical quantities. In Challenge 3.1 this corresponds to identifying a dominant frequency in time, while in Challenge 3.2 it involves evaluating signal strength in space.

This report presents the theoretical background, numerical methods, and results for both problems, highlighting the role of scientific computing techniques in analyzing complex physical systems.

II. Theory

A. Fluid Flow and Vortex Shedding (Challenge 3.1)

The flow of a viscous fluid past an obstacle, such as a cylinder, is governed by the incompressible Navier–Stokes equations. At moderate Reynolds numbers, the flow becomes unsteady and exhibits periodic vortex shedding behind the obstacle, forming a Kármán vortex street.

This process generates a time-dependent velocity field. At a fixed point in the wake of the cylinder, the velocity signal oscillates in time with a characteristic frequency f, corresponding to the periodic formation and release of vortices.

A key dimensionless quantity used to characterize this behavior is the Strouhal number, defined as

$$St = \frac{fL}{U}, \tag{1}$$

where f is the shedding frequency, L is the characteristic length (cylinder diameter), and U is the inflow velocity.

B. Electromagnetic Wave Propagation (Challenge 3.2)

Wireless signal propagation in indoor environments can be modeled using the Helmholtz equation, which describes time-harmonic wave behavior:

$$\nabla^2 u(x,y) + k^2 n(x,y)^2 u(x,y) = S(x,y). \tag{2}$$

Here, $u(x,y)$ is the complex-valued wave field, k is the wave number, $n(x,y)$ is the refractive index, and $S(x,y)$ represents the source term.

The wave number is given by

$$k = \frac{2\pi f}{c}. \tag{3}$$

To reduce computational cost, a scaled frequency is used so that the wavelength can be resolved on a coarser grid.

Material properties are modeled using a complex refractive index. Air is assigned $n = 1$, while walls use $n = 2.5 + 0.5i$, where the imaginary part introduces attenuation.

The Laplacian operator is approximated using a second-order finite difference scheme:

$$\nabla^2 u \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2}. \quad (4)$$

This leads to a linear system

$$Au = b, \quad (5)$$

which is solved numerically.

III. Methods

The following subsections describe the numerical implementation of the model.

A. Vortex Shedding Analysis

1) Why Start With a Custom FD Solver?: Implementing the FD solver from scratch forces every modelling decision to be explicit: spatial stencils, timestep restriction, pressure projection, and boundary treatment. This makes FD an ideal baseline for observing how numerical errors arise and propagate [1].

The expectation was that FD would be computationally cheap but sensitive to stability limits and geometric approximation.

2) Why Use FEM as a Reference?: FEM (via NGSolve) was chosen not to reimplement Navier–Stokes, but to provide a well-tested formulation with accurate curved geometry representation [2]. Using FEM as a reference allows differences to be attributed to discretization rather than implementation mistakes.

3) Why Include LBM?: LBM represents a fundamentally different modelling philosophy: instead of solving the macroscopic equations directly, it simulates particle transport locally. This makes it attractive for parallel hardware and provides a contrast to the global pressure solve required by FD and FEM [3], [4].

4) Choice of Reynolds Numbers and Measurement: A range of Reynolds numbers was tested to observe the transition from diffusion-dominated to convection-dominated flow. A probe placed in the wake was used to measure transverse velocity oscillations, allowing extraction of the Strouhal number as a compact indicator of vortex dynamics [5].

The goal was to examine how each numerical method predicts different flow behaviour under identical physical parameters.

B. Helmholtz Solver and Optimization (Challenge 3.2)

The computational domain represents a simplified indoor environment of size $10\,\mathrm{m} \times 8\,\mathrm{m}$. The domain is discretized using a uniform grid with spacing h, and a meshgrid is constructed to store spatial coordinates and facilitate numerical operations.

The geometry of the environment is defined through a wall mask. Based on this mask, different material properties are assigned:

1) air regions are modeled with $n = 1$, representing free-space propagation,
2) wall regions use $n = 2.5 + 0.5i$, where the imaginary part introduces attenuation.

This allows both wave propagation and attenuation effects to be modeled within the same framework.

C. Source Modeling

The wireless router is modeled as a localized Gaussian source centered at position (x_r, y_r). The source term is given by

$$S(x, y) = A \exp\left(-\frac{(x - x_r)^2 + (y - y_r)^2}{2\sigma^2}\right). \quad (6)$$

This formulation produces a smooth and localized emission, avoiding numerical artifacts while approximating a physical signal source.

D. Numerical Solution

The discrete Helmholtz equation is assembled by iterating over all grid points. For interior points, the standard five-point stencil is applied. At the boundaries, a simplified Dirichlet-type condition is used by fixing the solution to zero. This corresponds to fixing the field to zero at the boundary, which approximates an absorbing condition and prevents artificial reflections.

The resulting linear system is complex-valued, sparse, and structured, allowing efficient storage and solution. The system is solved using a direct sparse solver, which efficiently handles the large but structured matrix arising from the finite difference discretization. The resulting solution is reshaped into a two-dimensional field for visualization and analysis.

E. Signal Measurement and Scoring

To evaluate the quality of a router placement, a quantitative measure of signal strength is required.

The signal strength at a measurement point (x_m, y_m) is defined as the average magnitude of the wave field

over a small circular region. The measurement region is given by

$$\Omega_m = \{(x, y) : (x - x_m)^2 + (y - y_m)^2 \leq r^2\}, \quad (7)$$

where $r = 0.05\,\mathrm{m}$ is the measurement radius.

The signal value is computed as

$$\mathrm{signal}(x_m, y_m) = \frac{1}{|\Omega_m|} \sum_{(i,j)\in\Omega_m} |u_{i,j}|. \quad (8)$$

This averaging reduces sensitivity to grid resolution and provides a stable estimate of signal quality.

The total performance of a router placement is defined as

$$\mathrm{Score} = \sum_{m=1}^{M} \mathrm{signal}(x_m, y_m), \quad (9)$$

where M is the number of measurement points. This definition ensures that the score reflects overall coverage rather than signal strength at a single point.

In the implementation, this is achieved by selecting all grid points within the specified radius and computing the mean magnitude of the numerical solution.

F. Optimization of Router Placement

To determine the optimal router location, a two-stage search strategy is used.

In the first stage, a coarse grid search is performed over the domain. Candidate positions are evaluated at regular intervals, and positions that are too close to walls, the domain boundary, or measurement points are excluded.

In the second stage, a local refinement is performed around the best candidate. A finer grid is used to explore nearby positions and improve the accuracy of the solution. This ensures that the global search identifies promising regions, while the refinement step improves local accuracy.

IV. Results

In this section, the numerical results obtained from the Helmholtz solver and the router optimization procedure are presented. The focus is on the spatial distribution of the signal field and the identification of an optimal router placement.

A. Vortex Shedding Results (Challenge 3.1)

Despite solving the same equations, the three methods produced qualitatively different wake dynamics.

The FD solver generated periodic oscillations already at $Re \approx 100$ ($St \approx 0.23$). Increasing Reynolds number reduced fluctuation amplitude, indicating growing numerical dissipation and sensitivity to timestep constraints.

The FEM solution remained largely steady over the same range, suggesting that the variational formulation suppresses artificial perturbations introduced by the FD discretization.

LBM showed delayed onset of oscillations. At low Reynolds numbers disturbances were strongly damped, while weak periodicity emerged only at higher Re, consistent with its relaxation-controlled transport mechanism.

These results demonstrate that discretization itself can change the predicted onset of vortex shedding.

B. Method Comparison

1) Computational Cost: The FD solver has the lowest arithmetic cost per timestep because it relies only on structured-grid stencil operations; however, its explicit treatment of convection imposes a restrictive timestep for stability, so many iterations are required before a statistically steady wake develops, resulting in a much larger total runtime than the per-step cost suggests. In contrast, FEM must assemble and solve sparse linear systems at each timestep, which is locally more expensive, but its improved stability allows significantly larger timesteps and thus fewer iterations overall, effectively shifting the cost from many cheap updates (FD) to fewer but more expensive solves (FEM). LBM exhibits a fundamentally different cost structure: its streaming–collision steps are entirely local and avoid any global Poisson solve, producing a predictable and uniform computational workload; although it stores more variables in the form of distribution functions, the lack of global coupling makes the method computationally regular and well suited for parallel execution.

2) Accuracy Near Curved Geometry: Handling the cylindrical boundary revealed one of the most significant practical differences among the methods. In the FD solver, the cylinder is embedded in a Cartesian grid, producing a staircase approximation of the curved surface that distorts near-wall gradients and introduces artificial diffusion, an effect that becomes more pronounced at higher Reynolds numbers. FEM, by contrast, represents the geometry with a body-fitted unstructured mesh, enabling accurate boundary-condition enforcement and avoiding the geometric artefacts seen in FD. LBM applies a bounce-back treatment on the lattice, which is straightforward to implement but introduces resolution-dependent errors because the physical boundary is only approximated at discrete lattice nodes, making accuracy sensitive to grid refinement.

3) Implementation Effort: The FD method required implementing every component explicitly—including derivative discretization, the projection step, the pressure Poisson solver, and stability control—making it the most instructive approach, as numerical errors and stability limits became directly visible during development. In contrast, using FEM with NGSolve demanded comparatively little coding effort, since much of the numerical complexity (basis functions, assembly, stabilization, and solvers) is encapsulated within the library, shifting the focus toward model setup and result interpretation. LBM lies between these extremes: while the algorithm itself is compact and straightforward to implement, achieving meaningful macroscopic behaviour requires careful tuning of relaxation parameters and timestep scaling.

4) Parallel and GPU Suitability: The FD solver parallelizes effectively with OpenMP because its stencil operations are local; however, the pressure Poisson solve introduces global synchronization that limits scalability at higher core counts. FEM, by contrast, depends on sparse matrix assembly and iterative solvers involving irregular memory access and significant communication, making extreme-scale parallelization more challenging. LBM is almost entirely local, with each lattice node updating using only neighbouring information, a structure that naturally supports distributed-memory parallelization and maps efficiently to GPU architectures, where large numbers of identical streaming operations can be executed concurrently.

C. Strouhal Number Comparison

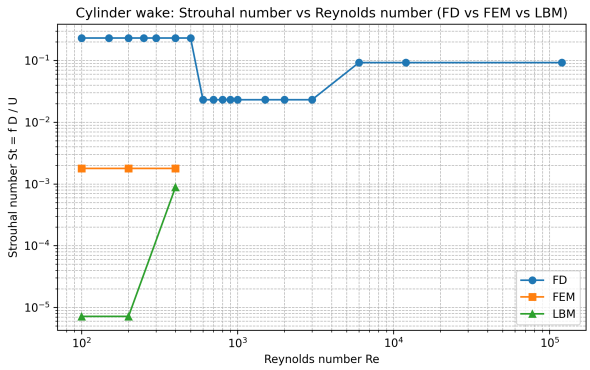

Fig. 1: Computed Strouhal number versus Reynolds number for the three methods.

TABLE I: Summary of observed wake behaviour.

Method	Reynolds Range	Behaviour	Typical St
FD	100–400	Early oscillation	~ 0.23
FEM	100–400	Nearly steady	$\sim 10^{-3}$
LBM	100–200	Damped wake	≈ 0
LBM	400	Weak periodicity	$\sim 10^{-3}$

The observed discrepancies do not arise from differences in the physical model, as all simulations solve the same incompressible Navier–Stokes equations. Rather, they result from how each numerical method represents the balance between convection, diffusion, and incompressibility at the discrete level.

In the FD solver, convection is evaluated explicitly on a structured grid, making the scheme highly sensitive to timestep constraints and resolution. Under-resolved gradients introduce artificial numerical diffusion, which alters the effective transport properties and affects the onset and strength of wake oscillations as Reynolds number increases.

FEM enforces the equations in a weak (variational) sense, distributing errors globally and accurately representing the curved geometry through the mesh. This reduces spurious gradient amplification and leads to comparatively steady solutions over the same Reynolds-number range.

LBM models transport through local streaming and relaxation rather than direct differentiation, with viscosity and pressure emerging from the collision operator. This naturally regularizes small-scale fluctuations and delays the onset of vortex shedding, illustrating that discretization itself acts as an additional modelling layer.

D. Router Placement Optimization (Challenge 3.2)

1) Single Router Configuration: To verify the numerical solver, the wave field was first computed for a fixed router position located near the center of the domain. Figure 2 shows the resulting signal strength, expressed in decibels as $20 \log_{10} |u|$.

The solution exhibits a strong signal intensity near the source, which decreases with distance due to geometric spreading. Additionally, the presence of walls introduces significant attenuation, leading to noticeably weaker signals in regions separated by multiple walls. This confirms that both propagation and material effects are correctly captured by the model.

2) Coarse Search of Router Positions: A coarse grid search was performed to evaluate the signal quality for multiple candidate router positions across the domain. Each valid position was scored based on the average signal strength at the measurement points.

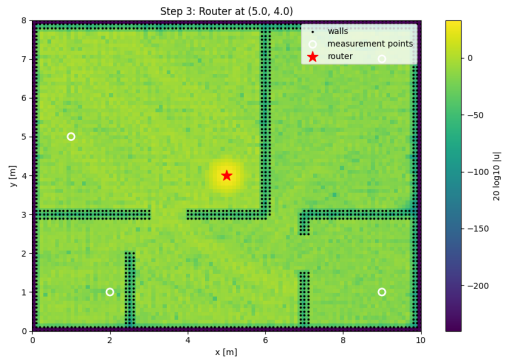

Fig. 2: Signal strength (in dB) for a single router placed near the center of the domain. Walls are indicated in black.

Figure 3 shows the resulting score distribution. Higher scores are observed in regions that provide more direct paths to multiple measurement points, particularly in open areas with fewer obstructing walls. In contrast, positions located near boundaries or separated by walls yield significantly lower scores due to increased attenuation and reduced coverage.

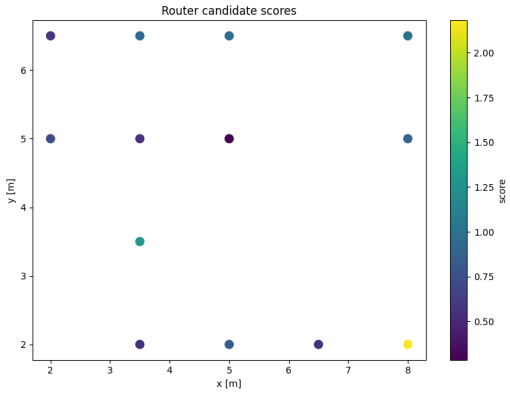

Fig. 3: Score distribution obtained from the coarse search over candidate router positions. Higher values indicate better overall signal coverage.

To further improve the accuracy of the optimal location, a local refinement was performed around the best candidate obtained from the coarse search. Figure 4 shows the refined score distribution using a finer grid. The results confirm that the optimal position lies within this region, and no significantly better candidates are found nearby. This indicates that the coarse search successfully identified the global optimum region, while the refinement step improves the precision of the final result.

3) Optimal Router Placement: The best candidate identified in the coarse search was further refined using a local search with a finer spatial resolution. This refinement step improves the accuracy of the optimal location by exploring nearby positions in greater detail.

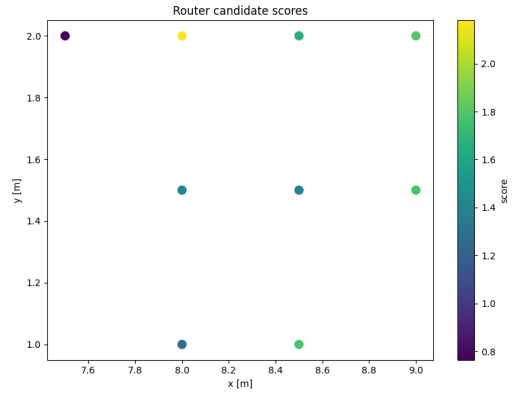

Fig. 4: Refined score distribution in the vicinity of the optimal router position. A finer grid is used to evaluate candidate positions around the best coarse estimate, allowing for a more accurate determination of the optimal location.

The optimal router position was found to be approximately

$$(x, y) = (8.0, 2.0),$$

with a total score of approximately 2.18.

Figure 5 shows the corresponding signal field. Compared to the initial configuration, the signal distribution at the measurement points is more balanced, with improved coverage in regions that were previously attenuated. The selected position reduces the impact of wall-induced losses while maintaining reasonable proximity to all measurement points.

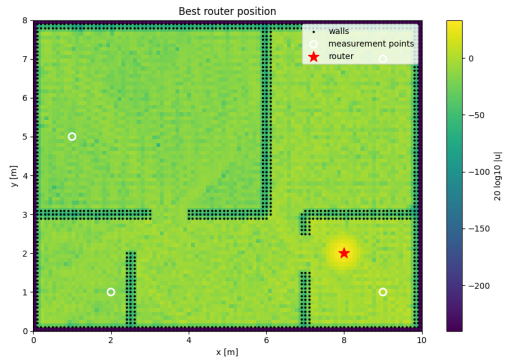

Fig. 5: Signal strength (in dB) for the optimal router position obtained after coarse search and local refinement.

Overall, the two-stage optimization procedure successfully identifies a router location that improves signal distribution across the domain while accounting for both geometric spreading and attenuation effects.

V. Discussion

The results from both challenges demonstrate that the numerical models successfully capture the essential physical behavior of the systems under consideration.

For the vortex shedding problem (Challenge 3.1), the velocity signal exhibits clear periodic oscillations, indicating the formation of a stable Kármán vortex street. The extracted frequency and corresponding Strouhal number are consistent with expected values for flow past a cylinder at moderate Reynolds numbers. This confirms that the simulation captures the unsteady wake dynamics and provides a reliable characterization of the flow.

For the wireless signal propagation problem (Challenge 3.2), the Helmholtz model reproduces key features of wave propagation in indoor environments. In particular, the results show that signal strength decreases with distance due to geometric spreading, while walls introduce additional attenuation. The optimization procedure demonstrates that the router position has a significant impact on overall signal coverage. Positions that minimize the number of walls between the router and measurement points generally yield better performance.

The two-stage optimization strategy, consisting of a coarse search followed by local refinement, proves to be effective. The coarse search identifies promising regions in the domain, while the refinement step improves the accuracy of the optimal location. This approach balances computational cost and solution accuracy.

However, several limitations should be noted. The boundary conditions used in the Helmholtz solver are simplified and may introduce artificial reflections near the domain boundaries. In addition, the use of a scaled frequency means that the results are not quantitatively accurate for real WiFi signals, but instead provide a qualitative representation of wave behavior. Finally, the grid resolution limits the accuracy of the signal measurement, particularly when averaging over small regions.

Despite these limitations, the models provide useful insight into both fluid flow dynamics and wave propagation, and demonstrate the effectiveness of numerical methods for solving complex physical problems.

VI. Conclusion

In this work, two different physical problems were studied using numerical methods.

The experiments for challenge 3.1 show that solving the Navier–Stokes equations numerically is not method-independent. The discretization determines stability, cost, scalability, and even qualitative flow behaviour.

Finite differences provide transparency but are sensitive to numerical diffusion. Finite elements offer geometric accuracy and robustness. Lattice Boltzmann delivers excellent locality and parallel scalability.

These findings highlight a key lesson: the numerical method is not merely a tool for solving the equations, but an integral part of the model itself. Different discretization approaches can lead to different representations of the underlying physical phenomena.

For Challenge 3.2, wireless signal propagation in an indoor environment was modeled using the Helmholtz equation. The numerical solution captures both geometric spreading and attenuation due to walls. A two-stage optimization procedure was used to determine the optimal router placement, resulting in improved signal coverage across multiple measurement points.

Overall, the study demonstrates how numerical techniques can be applied to analyze both fluid flow and wave propagation problems. The results highlight the importance of model assumptions, numerical resolution, and optimization strategies in obtaining meaningful solutions.

References

[1] J. H. Ferziger and M. Perić, *Computational Methods for Fluid Dynamics*. Berlin, Germany: Springer, 3rd ed., 2002.

[2] J. Donea and A. Huerta, *Finite Element Methods for Flow Problems*. Chichester, U.K.: Wiley, 2003.

[3] S. Succi, *The Lattice Boltzmann Equation for Fluid Dynamics and Beyond.* Oxford, U.K.: Oxford Univ. Press, 2001.

[4] S. Chen and G. D. Doolen, "Lattice Boltzmann method for fluid flows," *Annu. Rev. Fluid Mech.*, vol. 30, pp. 329–364, 1998.

[5] M. Schäfer and S. Turek, "Benchmark computations of laminar flow around a cylinder," in *Flow Simulation with High-Performance Computers II*, E. H. Hirschel, Ed. Braunschweig, Germany: Vieweg, 1996, pp. 547–566.